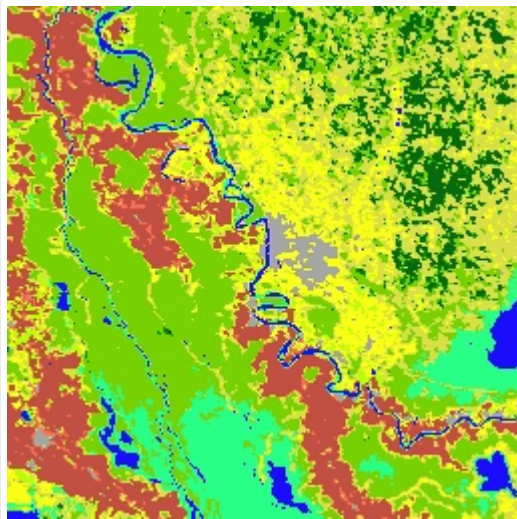


MCD12Q1.061 MODIS Land Cover Type Yearly Global 500m



◆ Page Summary



Dataset Availability

2001-01-01T00:00:00Z–2024-01-01T00:00:00Z

Dataset Producer

NASA LP DAAC at the USGS EROS Center
(<https://doi.org/10.5067/MODIS/MCD12Q1.061>)

Earth Engine Snippet

```
ee.ImageCollection(  
  "MODIS/061/MCD12Q1"  
)   
(https://code.earthengine.google.com/?scriptPath=Examples:Datasets/MODIS/MODIS\_061\_MCD12Q1)
```

Cadence

1 Year

Tags

landcover (/earth-engine/datasets/tags/landcover)

landuse-landcover (/earth-engine/datasets/tags/landuse-landcover)

modis (/earth-engine/datasets/tags/modis)

nasa (/earth-engine/datasets/tags/nasa)

usgs (/earth-engine/datasets/tags/usgs)

yearly (/earth-engine/datasets/tags/yearly)

Bands Pixel size: 500 meters (all bands)					
Name	Units	Min	Max	Pixel Size	Description
LC_Type1				500 meters	Land Cover Type 1: Annual International Geosphere-Biosphere Programme (IGBP) classification
LC_Type2				500 meters	Land Cover Type 2: Annual University of Maryland (UMD) classification
LC_Type3				500 meters	Land Cover Type 3: Annual Leaf Area Index (LAI) classification
LC_Type4				500 meters	Land Cover Type 4: Annual BIOME-Biogeochemical Cycles (BGC) classification
LC_Type5				500 meters	Land Cover Type 5: Annual Plant Functional Types classification
LC_Prop1_Assessment	%	0	100	500 meters	LCCS1 land cover layer confidence
LC_Prop2_Assessment	%	0	100	500 meters	LCCS2 land use layer confidence
LC_Prop3_Assessment	%	0	100	500 meters	LCCS3 surface hydrology layer confidence
LC_Prop1				500 meters	FAO-Land Cover Classification System 1 (LCCS1) land cover layer
LC_Prop2				500 meters	FAO-LCCS2 land use layer
LC_Prop3				500 meters	FAO-LCCS3 surface hydrology layer
QC				500 meters	Product quality flags
LW				500 meters	Binary land (class 2) / water (class 1) mask derived from MOD44W

LC_Type1 Class Table

Value	Color	Description
1	#05450a	Evergreen Needleleaf Forests: dominated by evergreen conifer trees (canopy >2m). Tree cover >60%.
2	#086a10	Evergreen Broadleaf Forests: dominated by evergreen broadleaf and palmate trees (canopy >2m). Tree cover >60%.
3	#54a708	Deciduous Needleleaf Forests: dominated by deciduous needleleaf (larch) trees (canopy >2m). Tree cover >60%.
4	#78d203	Deciduous Broadleaf Forests: dominated by deciduous broadleaf trees (canopy >2m). Tree cover >60%.
5	#009900	Mixed Forests: dominated by neither deciduous nor evergreen (40-60% of each) tree type (canopy >2m). Tree cover >60%.
6	#c6b044	Closed Shrublands: dominated by woody perennials (1-2m height) >60% cover.
7	#dcd159	Open Shrublands: dominated by woody perennials (1-2m height) 10-60% cover.
8	#dade48	Woody Savannas: tree cover 30-60% (canopy >2m).
9	#fbff13	Savannas: tree cover 10-30% (canopy >2m).
10	#b6ff05	Grasslands: dominated by herbaceous annuals (<2m).
11	#27ff87	Permanent Wetlands: permanently inundated lands with 30-60% water cover and >10% vegetated cover.
12	#c24f44	cropland.
13	#a5a5a5	Urban and Built-up Lands: at least 30% impervious surface area including building materials, asphalt and vehicles.
14	#ff6d4c	Cropland/Natural Vegetation Mosaics: mosaics of small-scale cultivation 40-60% with natural tree, shrub, or herbaceous vegetation.
15	#69fff8	Permanent Snow and Ice: at least 60% of area is covered by snow and ice for at least 10 months of the year.

16 #f9ffa4 (sand, rock, soil) areas with less than 10% vegetation.

17 #1c0dff Water Bodies: at least 60% of area is covered by permanent water bodies.

LC_Type2 Class Table

Value	Color	Description
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0	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.
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1	#05450a	Evergreen Needleleaf Forests: dominated by evergreen conifer trees (canopy >2m). Tree cover >60%.
---	---------	---

2	#086a10	Evergreen Broadleaf Forests: dominated by evergreen broadleaf and palmate trees (canopy >2m). Tree cover >60%.
---	---------	--

3	#54a708	Deciduous Needleleaf Forests: dominated by deciduous needleleaf (larch) trees (canopy >2m). Tree cover >60%.
---	---------	--

4	#78d203	Deciduous Broadleaf Forests: dominated by deciduous broadleaf trees (canopy >2m). Tree cover >60%.
---	---------	--

5	#009900	Mixed Forests: dominated by neither deciduous nor evergreen (40-60% of each) tree type (canopy >2m). Tree cover >60%.
---	---------	---

6	#c6b044	Closed Shrublands: dominated by woody perennials (1-2m height) >60% cover.
---	---------	--

7	#dcd159	Open Shrublands: dominated by woody perennials (1-2m height) 10-60% cover.
---	---------	--

8	#dade48	Woody Savannas: tree cover 30-60% (canopy >2m).
---	---------	---

9	#fbff13	Savannas: tree cover 10-30% (canopy >2m).
---	---------	---

10	#b6ff05	Grasslands: dominated by herbaceous annuals (<2m).
----	---------	--

11	#27ff87	Permanent Wetlands: permanently inundated lands with 30-60% water cover and >10% vegetated cover.
----	---------	---

12	#c24f44	cropland.
----	---------	-----------

13	#a5a5a5	Urban and Built-up Lands: at least 30% impervious surface area including building materials, asphalt and vehicles.
14	#ff6d4c	Cropland/Natural Vegetation Mosaics: mosaics of small-scale cultivation 40-60% with natural tree, shrub, or herbaceous vegetation.
15	#f9ffa4	Non-Vegetated Lands: at least 60% of area is non-vegetated barren (sand, rock, soil) or permanent snow and ice with less than 10% vegetation.
LC_Type3 Class Table		
ValueColorDescription		
0	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.
1	#b6ff05	Grasslands: dominated by herbaceous annuals (<2m) including cereal croplands.
2	#dcd159	Shrublands: shrub (1-2m) cover >10%.
3	#c24f44	Broadleaf Croplands: bominated by herbaceous annuals (<2m) that are cultivated with broadleaf crops.
4	#fbff13	Savannas: between 10-60% tree cover (>2m).
5	#086a10	Evergreen Broadleaf Forests: dominated by evergreen broadleaf and palmate trees (canopy >2m). Tree cover >60%.
6	#78d203	Deciduous Broadleaf Forests: dominated by deciduous broadleaf trees (canopy >2m). Tree cover >60%.
7	#05450a	Evergreen Needleleaf Forests: dominated by evergreen conifer trees (canopy >2m). Tree cover >60%.
8	#54a708	Deciduous Needleleaf Forests: dominated by deciduous needleleaf (larch) trees (canopy >2m). Tree cover >60%.
9	#f9ffa4	Non-Vegetated Lands: at least 60% of area is non-vegetated barren (sand, rock, soil) or permanent snow and ice with less than 10% vegetation.
10	#a5a5a5	Urban and Built-up Lands: at least 30% impervious surface area including building materials, asphalt and vehicles.
LC_Type4 Class Table		

Value	Color	Description
0	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.
1	#05450a	Evergreen Needleleaf Vegetation: dominated by evergreen conifer trees and shrubs (>1m). Woody vegetation cover >10%.
2	#086a10	Evergreen Broadleaf Vegetation: dominated by evergreen broadleaf and palmate trees and shrubs (>1m). Woody vegetation cover 10%.
3	#54a708	Deciduous Needleleaf Vegetation: dominated by deciduous needleleaf (larch) trees and shrubs (>1m). Woody vegetation cover 10%.
4	#78d203	Deciduous Broadleaf Vegetation: dominated by deciduous broadleaf trees and shrubs (>1m). Woody vegetation cover >10%.
5	#009900	Annual Broadleaf Vegetation: dominated by herbaceous annuals (<2m). At least 60% cultivated broadleaf crops.
6	#b6ff05	Annual Grass Vegetation: dominated by herbaceous annuals (<2m) including cereal croplands.
7	#f9ffa4	Non-Vegetated Lands: at least 60% of area is non-vegetated barren (sand, rock, soil) or permanent snow/ice with less than 10% vegetation.
8	#a5a5a5	Urban and Built-up Lands: at least 30% impervious surface area including building materials, asphalt, and vehicles.
LC_Type5 Class Table		
Value	Color	Description
0	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.

1	#05450a	Evergreen Needleleaf Trees: dominated by evergreen conifer trees (>2m). Tree cover >10%.
2	#086a10	Evergreen Broadleaf Trees: dominated by evergreen broadleaf and palmate trees (>2m). Tree cover >10%.
3	#54a708	Deciduous Needleleaf Trees: dominated by deciduous needleleaf (larch) trees (>2m). Tree cover >10%.
4	#78d203	Deciduous Broadleaf Trees: dominated by deciduous broadleaf trees (>2m). Tree cover >10%.
5	#dcd159	Shrub: Shrub (1-2m) cover >10%.
6	#b6ff05	not cultivated.
7	#dade48	Cereal Croplands: dominated by herbaceous annuals (<2m). At least 60% cultivated cereal crops.
8	#c24f44	Broadleaf Croplands: dominated by herbaceous annuals (<2m). At least 60% cultivated broadleaf crops.
9	#a5a5a5	Urban and Built-up Lands: at least 30% impervious surface area including building materials, asphalt, and vehicles.
10	#69fff8	Permanent Snow and Ice: at least 60% of area is covered by snow and ice for at least 10 months of the year.
11	#f9ffa4	Non-Vegetated Lands: at least 60% of area is non-vegetated barren (sand, rock, soil) with less than 10% vegetation.

LC_Prop1 Class Table

Value	Color	Description
1	#f9ffa4	(sand, rock, soil) or permanent snow/ice with less than 10% vegetation.
2	#69fff8	Permanent Snow and Ice: at least 60% of area is covered by snow and ice for at least 10 months of the year.
3	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.
11	#05450a	Evergreen Needleleaf Forests: dominated by evergreen conifer trees (>2m). Tree cover >60%.

12	#086a10	Evergreen Broadleaf Forests: dominated by evergreen broadleaf and palmate trees (>2m). Tree cover >60%.
13	#54a708	Deciduous Needleleaf Forests: dominated by deciduous needleleaf (larch) trees (>2m). Tree cover >60%.
14	#78d203	Deciduous Broadleaf Forests: dominated by deciduous broadleaf trees (>2m). Tree cover >60%.
15	#005a00	Mixed Broadleaf/Needleleaf Forests: co-dominated (40-60%) by broadleaf deciduous and evergreen needleleaf tree (>2m) types. Tree cover >60%.
16	#009900	Mixed Broadleaf Evergreen/Deciduous Forests: co-dominated (40-60%) by broadleaf evergreen and deciduous tree (>2m) types. Tree cover >60%.
21	#006c00	Open Forests: tree cover 30-60% (canopy >2m).
22	#00d000	Sparse Forests: tree cover 10-30% (canopy >2m).
31	#b6ff05	Dense Herbaceous: dominated by herbaceous annuals (<2m) at least 60% cover.
32	#98d604	Sparse Herbaceous: dominated by herbaceous annuals (<2m) 10-60% cover.
41	#dcd159	Dense Shrublands: dominated by woody perennials (1-2m) 60% cover.
42	#f1fb58	Shrubland/Grassland Mosaics: dominated by woody perennials (1-2m) 10-60% cover with dense herbaceous annual understory.
43	#fbee65	Sparse Shrublands: dominated by woody perennials (1-2m) 10-60% cover with minimal herbaceous understory.

LC_Prop2 Class Table

Value	Color	Description

1	#f9ffa4	(sand, rock, soil) or permanent snow/ice with less than 10% vegetation.
2	#69fff8	Permanent Snow and Ice: at least 60% of area is covered by snow and ice for at least 10 months of the year.
3	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.
9	#a5a5a5	Urban and Built-up Lands: at least 30% of area is made up of impervious surfaces including building materials, asphalt, and vehicles.
10	#003f00	Dense Forests: tree cover >60% (canopy >2m).
20	#006c00	Open Forests: tree cover 10-60% (canopy >2m).
25	#e3ff77	Forest/Cropland Mosaics: mosaics of small-scale cultivation 40-60% with >10% natural tree cover.
30	#b6ff05	Natural Herbaceous: dominated by herbaceous annuals (<2m). At least 10% cover.
35	#93ce04	Natural Herbaceous/Croplands Mosaics: mosaics of small-scale cultivation 40-60% with natural shrub or herbaceous vegetation.
36	#77a703	Herbaceous Croplands: dominated by herbaceous annuals (<2m). At least 60% cover. Cultivated fraction >60%.
40	#dcd159	Shrublands: shrub cover >60% (1-2m).

LC_Prop3 Class Table

Value	Color	Description
1	#f9ffa4	(sand, rock, soil) or permanent snow/ice with less than 10% vegetation.
2	#69fff8	Permanent Snow and Ice: at least 60% of area is covered by snow and ice for at least 10 months of the year.
3	#1c0dff	Water Bodies: at least 60% of area is covered by permanent water bodies.
10	#003f00	Dense Forests: tree cover >60% (canopy >2m).

20	#006c00	Open Forests: tree cover 10-60% (canopy >2m).
27	#72834a	Woody Wetlands: shrub and tree cover >10% (>1m). Permanently or seasonally inundated.
30	#b6ff05	Grasslands: dominated by herbaceous annuals (<2m) >10% cover.
40	#c6b044	Shrublands: shrub cover >60% (1-2m).
50	#3aba73	Herbaceous Wetlands: dominated by herbaceous annuals (<2m) >10% cover. Permanently or seasonally inundated.
51	#1e9db3	months of the year.

QC Class Table

ValueColorDescription

0	None	Classified land: has a classification label and is land according to the water mask.
1	None	Unclassified land: not classified because of missing data but land according to the water mask, labeled as barren.
2	None	Classified water: has a classification label and is water according to the water mask.
3	None	Unclassified water: not classified because of missing data but water according to the water mask.
4	None	Classified sea ice: classified as snow/ice but water mask says it is water and less than 100m elevation, switched to water.
5	None	Misclassified water: classified as water but water mask says it is land, switched to secondary label.
6	None	Omitted snow/ice: land according to the water mask that was classified as something other than snow but with a maximum annual temperature below 1°C, relabeled as snow/ice.
7	None	Misclassified snow/ice: land according to the water mask that was classified as snow but with a minimum annual temperature greater than 1°C, relabeled as barren.

8 NoneBackfilled label: missing label from stabilization, filled with the pre-stabilized result.

9 NoneForest type changed: climate-based change to forest class.

LW Class Table

Value	Color	Description
1	#1c0dff	Water
2	#f9ffa4	Land,

Explore with Earth Engine

Important: Earth Engine is a platform for petabyte-scale scientific analysis and visualization of geospatial datasets, both for public benefit and for business and government users. Earth Engine is free to use for research, education, and nonprofit use. To get started, please [register for Earth Engine access](https://console.cloud.google.com/earth-engine).
(<https://console.cloud.google.com/earth-engine>)

Code Editor (JavaScript)

(#code-editor-javascript-sample)

```
var dataset = ee.ImageCollection('MODIS/061/MCD12Q1');
var igbpLandCover = dataset.select('LC_Type1');
var igbpLandCoverVis = {
  min: 1.0,
  max: 17.0,
  palette: [
```

```
'05450a', '086a10', '54a708', '78d203', '009900', 'c6b044', 'dcd159',  
'dade48', 'fbff13', 'b6ff05', '27ff87', 'c24f44', 'a5a5a5', 'ff6d4c',  
'69fff8', 'f9ffa4', '1c0dff'  
],  
};  
Map.setCenter(6.746, 46.529, 6);  
Map.addLayer(igbpLandCover, igbpLandCoverVis, 'IGBP Land Cover');
```

Open in Code Editor (https://code.earthengine.google.com/?scriptPath=Examples:Datasets/MODIS/MODIS_061_MCD12Q1)